

WASP-43b

R-0642

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170402 · created 2026-07-25T19:16:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457845.6425 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.018 +/- 0.057
Transit depth (Rp/Rs)^2	0.032 +/- 0.21 [%]
Orbital Inclination (inc)	77.88 +/- 2.95
Airmass coefficient 1 (a1)	54.88 +/- 15.63
Airmass coefficient 2 (a2)	-3.0 +/- 1.11
Scatter in the residuals of the lightcurve fit is	28.17 %
Best Comparison Star	#3 - [573, 219]
Optimal Aperture	2.0
Optimal Annulus	9.13
Transit Duration (day)	0.0013 +/- 0.0025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.018 $\pm$ 0.057 vs 0.1594 (deviation 1.99 σ)
Duration fit vs published	0.0013 d vs 0.0512 d (deviation 16.01 σ)
Mid-time O-C	-0.5 min
Depth SNR	0.3
Residual scatter	28.17 %

Light curve

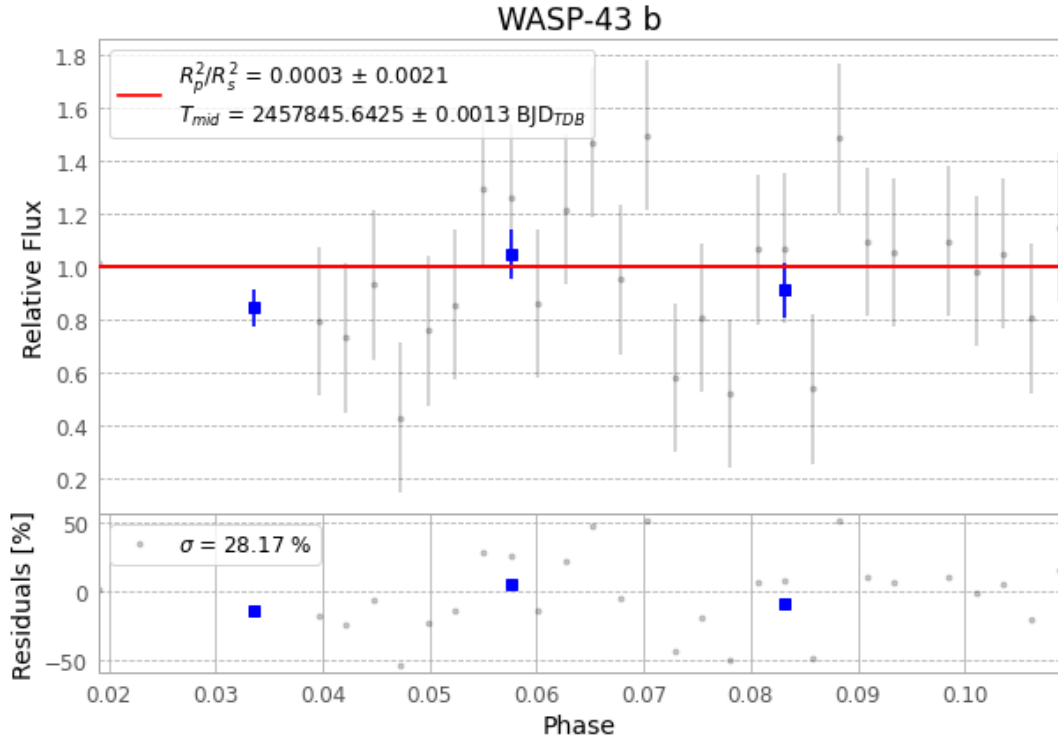


Figure 1 — FinalLightCurve_WASP-43 b_2017-04-01.png (SHA-256 7dfbff6bb507362a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 196], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3671b2c2bf7ead8b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

35 FITS frames (23 MB), WASP-43170402033912.FITS → WASP-43170402052412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170402.log (f763036dfce0...), FinalLightCurve_WASP-43 b_2017-04-01.png (7dfbff6bb507...), FinalLightCurve_WASP-43 b_2017-04-01.csv (3b4ace397057...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 19:16Z.